AP Precalculus — **Unit 1, Lesson 1.11**

Group Activity — Answer Key

Name: ____________________________ Partner: _______________________ Period: _________

Three Faces of the Same Function

A data analyst receives four rational functions, each expressed in a different form. For each function, the team needs to extract specific information. Sometimes the given form is ideal; sometimes you must convert to another form first. Your job is to identify what each form reveals and to perform the conversion when needed.

The four functions (each given in one form) are:

$$f(x) = \frac{x^2 - 2x - 8}{x^2 - x - 6} \qquad g(x) = \frac{x^3 + x^2 - 2x}{x - 1} \qquad h(x) = \frac{2(x+3)(x-1)}{(x+3)(x-4)} \qquad k(x) = \frac{x^2 + 5x + 6}{x + 1}$$

Tier R — Remediate

The numerator and denominator of f and h are already in factored form below. Fill in the table using the information each form directly gives you.

Fn	Factored form	Zeros (from factored form)	Hole or VA at each zero?
f	$\frac{(x-4)(x+2)}{(x-3)(x+2)}$	$x = 4,\ x = -2$	**hole at** $x = -2$**; VA at** $x = 3$
h	$\frac{2(x+3)(x-1)}{(x+3)(x-4)}$	$x = -3,\ x = 1$	**hole at** $x = -3$**; VA at** $x = 4$

For each function, circle the form that best reveals end behavior: f: **Factored** / **Standard** h: **Factored** / **Standard**

Tier A — Apply

1. Convert $f(x)$ from its given form to factored form. Identify all zeros, holes, and VAs.

   Factored f: $\dfrac{(x-4)(x+2)}{(x-3)(x+2)}$

   Zeros: $x = 4$ Hole(s): $(-2, 6)$ VA(s): $x = 3$

2. For $g(x)$, the degree of the numerator is one more than the degree of the denominator. Use polynomial long division to rewrite $g(x)$ as $q(x) + \dfrac{r}{x-1}$.

   Long division work space:

   Factor numerator: $x^3 + x^2 - 2x = x(x^2 + x - 2) = x(x+2)(x-1)$.

   So $g(x) = \dfrac{x(x+2)(x-1)}{x-1} = x(x+2) = x^2 + 2x$ for $x \neq 1$.

   $g(x) = x^2 + 2x$ (remainder $= 0$, hole at $x = 1$, $y = 3$)

3. For $h(x)$, determine the end behavior using the standard-form leading terms. State the horizontal asymptote (or explain why there is none).

   Numerator leading term: $2x^2$ Denominator leading term: x^2

   Horizontal asymptote: $y = 2$ or Slant/end behavior: **HA at $y = 2$; same degree**

Tier E — Extend

1. Use polynomial long division on $k(x) = \dfrac{x^2 + 5x + 6}{x+1}$. Write $k(x) = q(x) + \dfrac{r}{x+1}$ and state the equation of the slant asymptote.

   Dividing: $x^2 + 5x + 6 \div (x+1)$: quotient $x + 4$, remainder 2.

   $k(x) = x + 4 + \dfrac{2}{x+1}$ Slant asymptote: $y = x + 4$

2. Explain in a complete sentence how you would use the result of long division to determine whether $k(x)$ has a slant asymptote or a horizontal asymptote.

   If the degree of the numerator is exactly one more than the degree of the denominator, the quotient after long division is a non-constant linear polynomial, giving a slant asymptote $y = q(x)$. If degrees are equal, the quotient is a constant, giving a horizontal asymptote. . . . . .

3. Construct a rational function $p(x)$ that satisfies *all* of the following: a hole at $x = 2$, a vertical asymptote at $x = -1$, zeros at $x = 0$ and $x = 5$, and a slant asymptote. Write p in factored form and verify the degree condition for the slant asymptote.

   $p(x) = \dfrac{x(x-5)(x-2)}{(x+1)(x-2)}$ (many valid answers)

   Degree of numerator: **3** Degree of denominator: **2** Difference: **1**

Teacher Note

Tier R: Students should recognize that factored form directly gives zeros and hole/VA classification without additional work.
Tier A item 2: The numerator factors cleanly, so no numeric long division is needed. Accept either the factoring approach or numeric long division.
Tier E item 3: Any rational function with numerator degree = denominator degree +1, the correct zeros, hole at $x = 2$, and VA at $x = -1$ is acceptable. The example shown is the minimal such construction.